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AI and Security, from an information security and risk manager standpoint

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ABSTRACT Fields such as machine learning and artificial intelligence are proven business enablers, we have several use cases in the field of technology from basic implementations such as automated scanners, and weak forms such as Alexa and Siri that have paved the way for further development. Search for more avenues of investment in this field continues till date. It's also evident that researchers and organizations have barely scratched the surface in terms of exploring and using AI (Artificial intelligence) as a technology. Theoretical forms of AI have not yet been implemented. This paper aims to provide some context around the various forms of AI currently used and can be explored in the near future. It's important to be mindful that, with the use of any technology, even AI, there are challenges, regulations, and security risks that tag along. Firms need to pay close attention to the macro factors that could impact the adoption and AI in their respective businesses. This study aims to provide a brief overview of the aforementioned road blocks that could impact AI adoption, however, there are some solution strategies or approaches that could help firms along the way to overcome those road blocks. These strategies might take time to build and take effect but they will be beneficial in the long run and thus help with not only sustainable implementation, but also complete and successful adoption of AI as a technology enabler.

INDEX TERMS Artificial intelligence, Generative AI, Security, Risk, Risk management, AI for security, Security for AI, AI security

I. INTRODUCTION

Driving towards automation and automating manual processes and tools have always ignited innovation, the invention of new technologies, be it the invention of wheels to ease the load of manually carrying materials, advancing agriculture methods, or the invention of computer systems to undertake complex transactions [1]. The need for systems and machines was always to be an enabler and make our lives easier, in some cases to achieve things that would have felt impossible, for example, the advancements in the aviation sector, exploring space missions, etc. However, there was always this existential question of whether machines would have a thinking of their own and is there more to them. The question pondered by none other than the great "Alan Turing", computer scientist and mathematician, he did vision a future where machines would perform certain tasks better than humans [2]. We are living in a time where researchers experience ongoing debates on the meaning of intelligence [3] and Turing's vision of thinking machines is getting closer each day [4]. At the moment Generative AI is where we are [5], [6].

There are several definitions coined for AI over the years, as per the United States government and for legislation, the

following definition is worded 'artificial intelligence' means a machine-based system that can, for a given set of human-defined objectives, make predictions, recommendations or decisions influencing real or virtual environments [7]. Artificial intelligence is a machine's ability to perform cognitive functions that are usually associated with the human minds [1]. Over a sustained period, businesses and governments have invested in AI, which has resulted in advancements in technologies such as mapping technologies and smart phones. Businesses have consistently tried to adopt AI in their product features such as Siri and Alexa, which try to understand the input and come up with outputs that match user expectations. AI, in its simplest form, combines computer science concepts and robust datasets involving location, behavior, pattern etc., to come up with solutions [7].

With the rapid pace of development at scale, in the field of AI and investments matching those efforts, it is important to understand the risks and how to be both proactive and reactive in managing them. It is in everyone's interests, especially businesses and governments, to ensure that the risks stemming from this domain have been guided through an appropriate life cycle and that the subject matter experts

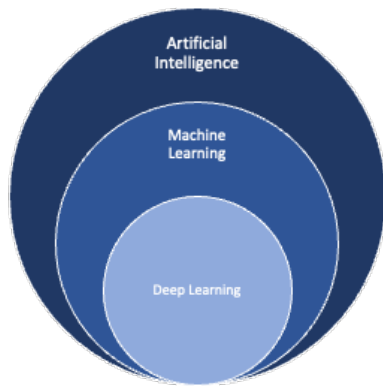


FIGURE 1. Artificial intelligence vs Machine learning vs Deep learning

are involved. This would be one of the measures that helps advance technology such as AI, to have a sustainable future.

This paper tries to connect the fields of AI, information security and risk, it also tries to shed light on the various nuances in this interconnection, through the following sections, furthermore, it aims to answer questions such as "How can AI be used in Security?", "How can Security concepts can be implemented in the AI technologies and tools?". This study not only tries to point out the risks and challenges involved in the implementation of AI, but also recommends solutions to the afore-mentioned challenges.

II. AI FOR SECURITY VS SECURITY FOR AI

AI is a complex information system that is capable of consuming enormous amount of data and simulates human decision making, not thinking. AI should not be confused with Business intelligence and related tools. AI as a technology is still in the development phase because it requires human intervention and cannot independently function on its own. It has not reached the level of self-awareness, which is the highest form of AI. The following is interesting because sub-domains in AI, mainly machine learning and deep learning, have gained prominence in the field of research, and there has been significant amount of research done on the statistical models and algorithms that could be used. The weak form of AI or Narrow AI as its alternatively called has been existed for a few years now [8] [11].

As we tried to understand the above visual in [12, Fig. 1], Artificial intelligence (AI) is a means of solving complex problems without human intervention, and decisions would closely replicate humans, machine learning is a subset of AI that uses mathematical and statistical models to process raw data and generate information that can be leveraged [13]. Deep learning is a subset of machine learning inspired by our brain's network model, which comprises of neural network of more than three layers including the input and output layers. It can take data in its raw form, determine hierarchy, differentiate based on categories, use an algorithm and produce an output; it does not require human intervention. Machine learning requires human intervention and does not scale on

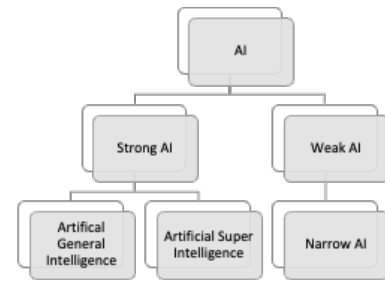


FIGURE 2. Weak AI vs Strong AI

its own like deep learning. This approach is based on training a model using data sets [8].

Generative AI, generates content in response to a prompt, the form of AI that is currently popular using deep learning models that can take raw data and produce statistically probable outputs. Generative models have been used for years for numerical analysis, and the use of deep learning concepts has enabled development of tools such as ChatGpt [1]. Uses: Speech recognition – we might have seen it with Siri and Alexa which is based on voice detection, devices can be managed and customer service – online virtual agents are replacing human agents responsible for basic questions, connecting us to humans only if needed, thus reducing the workload.

A. WEAK AI VS STRONG AI:

In [8], conceptually, there are two main variations in AI, as shown in [8, Fig. 2], there is Weak AI alternatively called Narrow AI, which has been trained to perform limited tasks. This model has been around for a few years now, and we have seen it as Alexa, Siri, etc. These tools listen to our commands as inputs and generate an output that may or may not match our expectations. Autonomous vehicles also use the same concept to an extent but improvements need to be made. Process AI, such as Jira automation cannot be explicitly type casted under weak AI, which is a form of workflow automation that takes in an input, follows a rule set and produces repeatable outcomes.

Strong AI is made up of Artificial general intelligence, where machines or systems could think like humans, have a self-aware consciousness, ability to solve problems, and so on, while artificial intelligence can surpass the ability of the human brain. Strong AI and its branches are theoretical concepts at the moment and not practical, do not have real world use cases at the moment. Researchers are currently exploring this possibility [8].

B. AI FOR SECURITY

"AI for Security" means how can we use AI technologies to solve security problems or use AI technology in security products or solutions. In [14], security product landscape can be split across the following categories:- AI can be embedded across these categories to varying degrees, depending on the technology, its use, and what is permitted or allowed as per

laws and regulations;

- 1) Observing : Surveillance systems would all fall under this category, where it is just meant for observing the environment, infrastructure, etc.
- 2) Detecting : Products like Intrusion detection systems (IDS), placed on perimeters, entry sensors, sonars, radars etc., fall under this category because they actively try to trace the presence of someone or something.
- 3) Controlling : Access control systems, firewalls, etc. fall in this category, where access is only allowed based on a pre-filtered, controlled list, policies, and nobody else can access the systems or information.
- 4) Responding : Intrusion prevention systems, weaponry, drones, robotics, etc. fall in this category, where the intention here is to respond to threats.

In [13], the field of information security, there is a need to collect large amounts of data, analyze it, and deploy control measures based on the pattern observed. These are leveraged in the following areas: malware detection, incident detection, anomalies, fraud detection, spam filters, user biometric authentication, and credit ratings. The algorithmic approach here is important, to keep up with threats and attacks, Data loss prevention (DLP) and Security incident and event management (SIEM) systems constantly need to be adapted and modified, which results in a huge strain in resources and time, deploying AI would help in efficiency in terms of time and resources. AI can be used in the sample selection of risk assessments, and we can train the model on historic data and patterns across teams. We would need qualitative information from stakeholders across teams, organization context based on financial risk appetite, and based on the repository of risks, vulnerabilities and incidents. We can employ machine learning algorithms to determine which type of attack frequent our products and firm, which type of vulnerability is a common occurrence, in which teams have a high probability of non-compliance [15].

Use cases: Apple uses AI in facial recognition, infrared sensors coupled with neural network technology, apple states that it is very difficult to cheat this technology and the model adapts according to changes in the user's facial features, including wearing a hat, glasses, etc. Since 2006, Qrator labs have been using a threat detection model to protect themselves against Distributed denial of service (DDOS) attacks based on the pattern learned from the visitors using the website, which model allows filtering attacks with a very high accuracy [13].

C. SECURITY FOR AI OR SECURITY IN AI:

“Security for AI” means designing, implementing security measures for AI technologies, since every technology has limitations, vulnerabilities and prone to attacks so it is imperative to secure AI infrastructure especially if its crucial to the business and nation state infrastructure. The next few sections will try to draw out the serious and major repercussions or risks that might result from cyber attacks.

1) Challenges associated with implementation of AI [6]

- 1) Blind trust in AI : The intention to adopt AI in automating operations is to free up critical resources and time of staff members and reduce manual interventions to enable efficiency; however, over reliance on AI can lead to false positives, blind trust, and lack of transparency meaning that if the models are not regularly looked at, and inspected for inaccuracies, it can lead to meaningless outputs. Furthermore, if the data inputs are no longer relevant, then the output wont serve the purpose, If we stayed away from the model and output for too long, we wont be able to figure out inaccuracies since model self corrects itself and keeps maturing over a period of time. It will be difficult to identify where it went wrong and what the contributing input or factor was.
- 2) Ethical challenges : Businesses need to be mindful of the data privacy regulations in the region, state, and nation they are operating since there are various regulatory considerations for AI, especially laws such as the AI bill of rights, the EU AI Act and others, GDPR (General Data Privacy regulation), and similar privacy regulations apply inherently [23] [24] [25]. That is a given because information being collected might fall into Personally Identifiable information (PII) / Material Nonpublic information (MNPI) categories within the scope of the regulations, for example, – biometric information, facial information, etc..
- 3) Data retention for some AI models[18] : Some AI solutions retain data for extended periods of time so that the model continues to learn and evolve , keep referencing it as part of its historic data-set, which increases the risk of unauthorized access . It is challenging to delete the data when the user requests it because of the nature and complexity of AI solutions.
- 4) Limited contextual insight : AI solutions rely on datasets that they are trained in and the models that have been built based on algorithms, its constantly evolving and learning based on these pre-set factors or lets call it pre-defined rules. During the Ransomware attack, certain firms employing AI had all their traffic blocked, based on a pre-defined set of rules, anomaly detection and training data set used. Because legitimate traffic was blocked, there was a significant business impact and downtime. In some cases, there might be unseen, inherent data bias risk, based on the data sets used as input to the model, for example, the Amazon applicant scandal in 2018, where AI was used to screen candidates, apparently, there was a gender bias found based on the review [21].
- 5) Lack of transparency : It is very difficult to determine why a certain decision was made, since the learning models would continuously evolve , making it difficult for researchers to investigate the rationale behind the decisions [14] , especially if the system employs a

deep learning concept with multiple layer of neural networks.

2) Risks resulting from use of AI [17]

- 1) Text based AI – This form of AI, especially ChatGpt, can be used to write code by attackers, generating scripts, that are not as sophisticated as Code based AI
- 2) Media based AI – This form of AI has been used on a massive scale in social media, deepfake videos, and audio being simulated using AI has resulted in an increase in fraud related attacks.
- 3) Code based AI – Even inexperienced attackers can write code that can hide malware, these malicious codes have the ability to bypass security controls.
- 4) Combination based AI – Attackers can use Media based AI to generate deep fake videos or something that can mimic a human identity and can combine it with a text based AI such as ChatGpt to write an email as a form of a spear-phishing attack or targeted attack, these attacks can result in identity theft, bank related fraud and many more since it's a complex form of attack driven by AI, Investigating these attacks and finding the perpetrator could be really challenging.

3) Risks resulting from implementation of AI

Implementing AI related solutions and technologies can result in cybersecurity risks [18]

- 1) Data breaches and misuse [18] : AI models rely on training data sets and inputs, as a result, these solutions, if not protected well could result in vulnerabilities that can be exploited by attackers. There might be various forms of data used, including PII/MNPI, which makes the target more attractive for a hacker, and there is an additional incentive. The risk multiples if the data used in these models are used in staging and development environments for daily use and testing, these lower environments do not have stringent controls. There might be accidental data breaches and misuse of data.
- 2) Data poisoning attack [19] : An attacker can intentionally spam data storage, insert incorrectly labeled data sets, or irrelevant data sets deviating from the median range of the training set, resulting in inaccurate predictions, insecure, and unreliable outcomes.
- 3) Model poisoning attack [18] : Adversarial attacks target AI models or systems in their production environment, whereas model poisoning attacks target AI models in a development or testing environment. In model poisoning, attackers introduce malicious data into the training data to influence the output — sometimes creating a significant deviation in behavior from the AI model.
- 4) Model inversion attack [19] : An attacker can reverse engineer the model, thus exposing the data set being used, resulting in access to PII/MNPI; if used, these attacks fly below the radar and are not easy to identify.
- 5) Output integrity attack [19] : Attackers can intentionally falsify the output through Man in the middle attack,

change the appearance of the output, and these attacks result in business impact because the output is not desired and false information that the business would rely on. This can have an impact on the business outcomes.

- 6) Vulnerabilities in AI infrastructure [18] : Like any software solution or technology, AI implementation requires software components and hardware, any vulnerabilities in proprietary components or 3rd party components used can result in exposure of the relevant piece in the AI infrastructure, and because of the complexity of the solution and infrastructure . Identifying the root cause is a challenging task. Attacks targeting these vulnerabilities might go undetected if there is no deviation in the output. These attacks also pivot to lateral attacks to provide greater access to the operating environment.

There are very limited strategies and product firms in the business of providing security solutions for AI technology. MITRE, a prominent and not for profit security, has built a framework specifically for AI called MITRE ATLAS™ (Adversarial Threat Landscape for Artificial-intelligence Systems) to raise awareness of the evolving vulnerabilities and tactics, techniques and procedures to tackle them [22]. The following section on the method and discussion will discuss in detail the various strategies and techniques that can be employed to tackle security risks and issues related to AI.

III. METHOD, DISCUSSION AND EXTENDED USE CASES

Considering the various challenges, risks in this space, and the business impact resulting from those risks, it is important to have a planned framework to tackle these roadblocks.

For overall security maturity related;

- 1) Responsible usage and culture : Since AI technologies are very simple to use, it is very important for businesses to educate staff and be mindful of AI usage reasons. what is the end goal, should be within the boundaries of laws and regulations, ensuring there is no harmful intent behind the development and usage of AI model(s).
Users and customers should be given complete autonomy when their personal data are used for training data sets. The culture must be driven from top to bottom, starting with the management to staff members.
- 2) Be mindful of the governance around AI : There are various regulations to be mindful of while designing AI solutions and products, governments have passed laws for example the White house – Bill of rights outlined that ensure its citizens are protected from unsafe, unethical versions of AI and always have the civil liberties promised to them [23].
Similarly, European Union has passed the EU AI Act; which aims to ensure that fundamental rights, democracy, the rule of law and environmental sustainability are protected from high risk AI, while boosting innovation and making Europe a leader in the field. The rules establish obligations for AI based on its potential risks and level of the impact [24].

UK NCSC (UK National Cybersecurity Centre)

The NCSC; UK version of CISA (US), continues to work with international partners and industry to provide guidance on the secure development and use of AI, and has released a set of guidelines on systems have been created from scratch or built on top of tools and services provided by others [26].

Organizations and businesses will have their own set of policies, standards and frameworks around AI, and teams need to be mindful of all of these internal governance structure in place

- 3) Third party contracts and reviews : Review contracts and contractual verbiage with vendors, ensure the agreements make it clear and align with the organizations overarching policies and procedures, ensure accountability, there are chances of vendors and 3rd parties not complying, be very careful around this de-lineation since they might be processors of information collected. Customers and users must be notified well in advance of these contractual obligations. Terminate contracts and find appropriate vendors if the risks outweigh business objectives. Frequently audit 3rd parties in case of AI related dependencies.
- 4) Core Security and Architecture : Invest in offensive security testing of the AI infrastructure, including the models and the data sets in use, ensure that the Red teams are involved early in the development phase, testing for architecture gaps, vulnerabilities in approach, and model recommendations. Suggestions to engineering teams for improvements.
- 5) Defensive security measures : Blue teams such as Security operations mainly vulnerability management, and incident management teams should also be on the lookout for anomalies in code, network, output from the models, deviation from median, patches, and vulnerabilities should be fixed immediately with accelerated Service level agreements (SLAs) [17]. Investments should also be made in the field of threat detection technologies like Extended detection and response (XDR), Data loss prevention (DLP) solutions.
- 6) Exclusive privacy related considerations : Organizations should be mindful of the type of data collected, duration, retention policies, and purposes. This information should be explicitly clear to consumers , customers, and users. Consent, transparency and autonomy are key concepts in privacy paradigm. Usage limitation and purpose specification made clear to users who have agreed to share data. Fairness – ensuring users have the right to view, edit, amend, delete data as they see fit and its used for ethical purposes based on their knowledge. The aforementioned regulations such as the US AI bill of rights, EU AI Act and GDPR, provide some very good guidelines in terms of privacy [23] [24] [25].

Risk management approach – The Mediator & Collaborator approach that can bring teams together in tackling risks



FIGURE 3. AI RMF - NIST

from and to AI. AI risk management can drive responsible uses and practices by prompting organizations and their internal teams who design, develop, and deploy AI to think more critically about the context and potential or unexpected negative and positive impacts [28].

While the Risk management framework (RMF) generally addresses negative impacts, it aims to minimize the anticipated negative impacts, identify opportunities to maximize the positive impact [27].

The AI RMF Core provides outcomes and actions that enable dialogue, understanding, and activities to manage AI risks and responsibly develop trustworthy AI systems.

As seen in [27, Fig.3] there are following sections in the AI RMF Core:

- GOVERN ensures strong governance that, in turn can drive and enhance internal practices and norms to facilitate an organizational risk culture.
- The MAP function establishes the context to frame risks related to an AI system.
- The MEASURE function employs quantitative, qualitative, or mixed-method tools, techniques, and methodologies to analyze, assess, benchmark, and monitor AI risks and related impacts.
- The MANAGE function allocates risk resources to mapped and measured risks on a regular basis, as defined by the GOVERN function.

There is an additional concept of AI RMF profiles, which are implementations of AI RMF functions, the current state can be evaluated and assigned a profile, while a temporal profile can be created for the target state, which indicates the outcomes needed to achieve the desired or target AI risk management goals. The RMF framework has various categories and sub categories, explained in detail, that can be leveraged for risk management purposes.

The risks identified through various avenues as listed in this section, can be analyzed, constantly tracked , monitored, and reported in the form of dashboards by risk management teams. Risk management teams can be plugged into various stakeholder and engineering teams to collaborate on risk iden-

tification avenues [16]. While the accountable teams work on remediation measures, risk teams can be on back-end, tracking monitoring for SLA breaches and escalations. The selected Subject matter experts (SMEs) from each of the stakeholders involved including the risk team, can decide on the holistic governance flow and structure, ensuring all of the risks have been through the risk management lifecycle of identification, analysis, response and reporting to ensure management oversight including Chief Information Security Officer (CISO) review.

The process can be leveraged regardless of the business sector in which the firm operates, and the size of the organization. To make this approach successful, teams need to have an agile mindset, move away from siloes to an inter-team collaboration model and reasonable power handed to the security teams to make data-driven decisions.

IV. CONCLUSION

The end goal of AI as a technology is to make human lives easier by reducing manual interventions and automating tasks. Automation should result in more efficient predictable outcomes and takes less time. As we have seen and read in this paper, AI is a very broad field with depth, there are various areas that are yet to be explored practically, only available as theoretical concepts at the moment, with the pace at which advancements expected in this field, super intelligent forms of AI will become a reality. AI is used in several technologies and use cases, it has multiple uses in the field of security as well, which have been demonstrated by businesses. At the same time, it is important to bear in mind the overarching regulations prescribed around the use of AI, privacy requirements, etc. There are many inherent risks surrounding this field, and the devastating consequences of AI misuse.

The analysis and results derived from this paper can be used to improve the security posture of firms considering of deploying AI in their environment or creating products driven by AI technologies. Risk management teams can play a pivotal role in supporting engineering and product teams that can help businesses experience the benefits of technology, while at the same time ensuring collaboration amongst various security teams, to draw out the inherent risks and work towards mitigating if not all, majority of those risks identified.

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